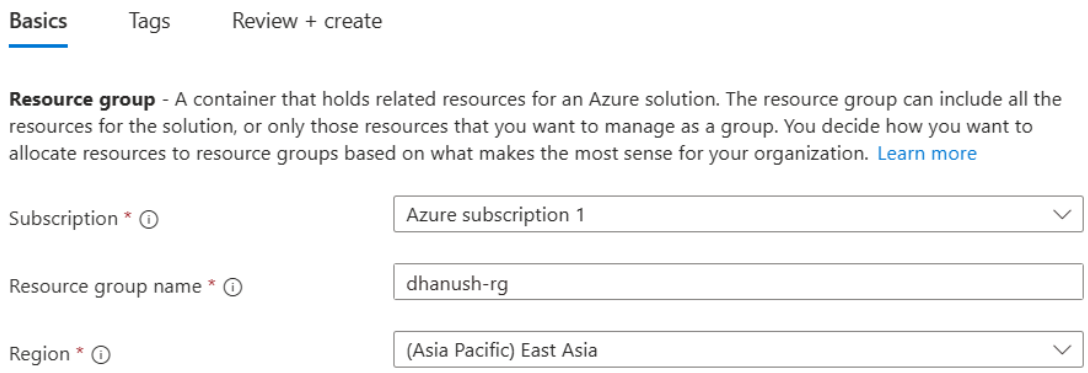
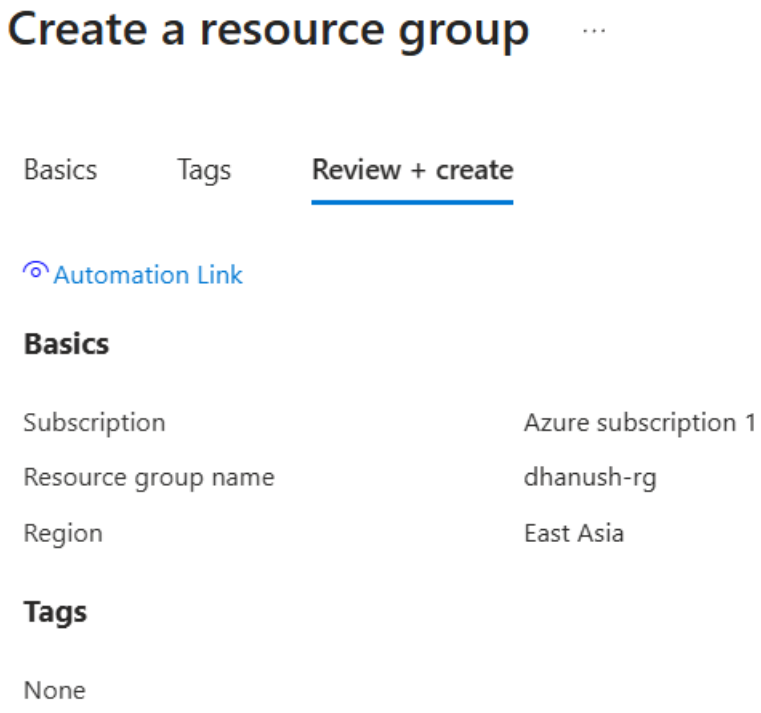
LAB 2 : Webserver VM

Create an Azure Linux VM and configure it as a webserver with Apache HTTPD and a sample HTML page

STEP 1 : Resource Group

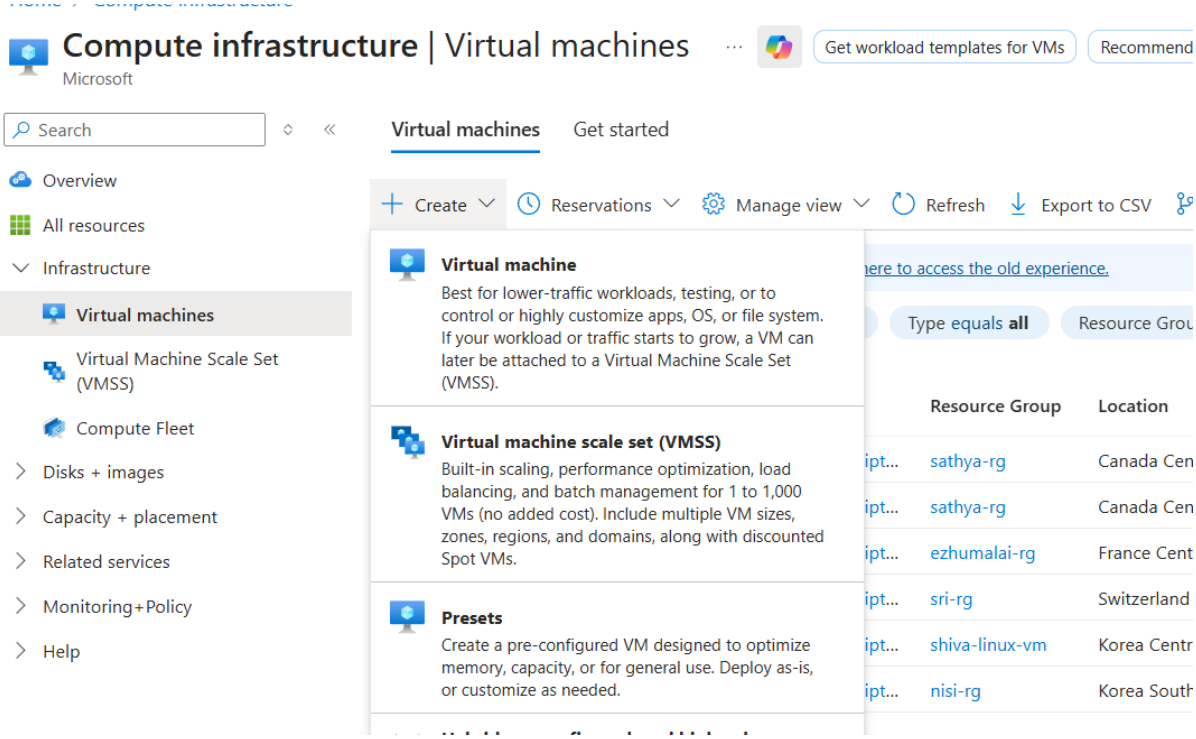




STEP 2 :Virtual Machine

Go to Virtual machines → Create → Azure Virtual Machine

- **Resource group: <studentname>-rg**
- **VM name: <studentname>-webserver**
- **Region: Select any region**
- **Image: AlmaLinux OS 9 – x64 Gen2**
- **Size: Standard_B1s (free tier eligible) (OR) Standard_D2s_v3**
- **Authentication: SSH public key (generate new or use existing)**
- **Inbound ports: Select SSH (22) and HTTP (80)**
- **Click Review + Create → Create**



Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ — Azure subscription 1

Resource group * ⓘ — dhanush-rg

Create new

Instance details

Virtual machine name * ⓘ — dhanush-webserver

Region * ⓘ — (Europe) Switzerland North

Deploy to an Azure Extended Zone

Availability options ⓘ — Availability zone

Zone options ⓘ

- (●) Self-selected zone
  Choose up to 3 availability zones, one VM per zone
- ( ) Azure-selected zone (Preview)
  Let Azure assign the best zone for your needs

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Availability options ⓘ: Availability zone

Zone options ⓘ:

- (●) Self-selected zone
  Choose up to 3 availability zones, one VM per zone
- ( ) Azure-selected zone (Preview)
  Let Azure assign the best zone for your needs

Availability zone * ⓘ: Zone 1

You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more

Security type ⓘ: Trusted launch virtual machines

Configure security features

Image * ⓘ: AlmaLinux OS 9 - x64 Gen2 (free services eligible)

See all images | Configure VM generation

VM architecture ⓘ:

- ( ) Arm64
- (●) x64

Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ: [ ]

Administrator account

Authentication type ⓘ:

- ( ) SSH public key
- (●) Password

Username * ⓘ: azureuser

Password *: ••••••••••••••

Confirm password *: ••••••••••••••

Inbound port rules

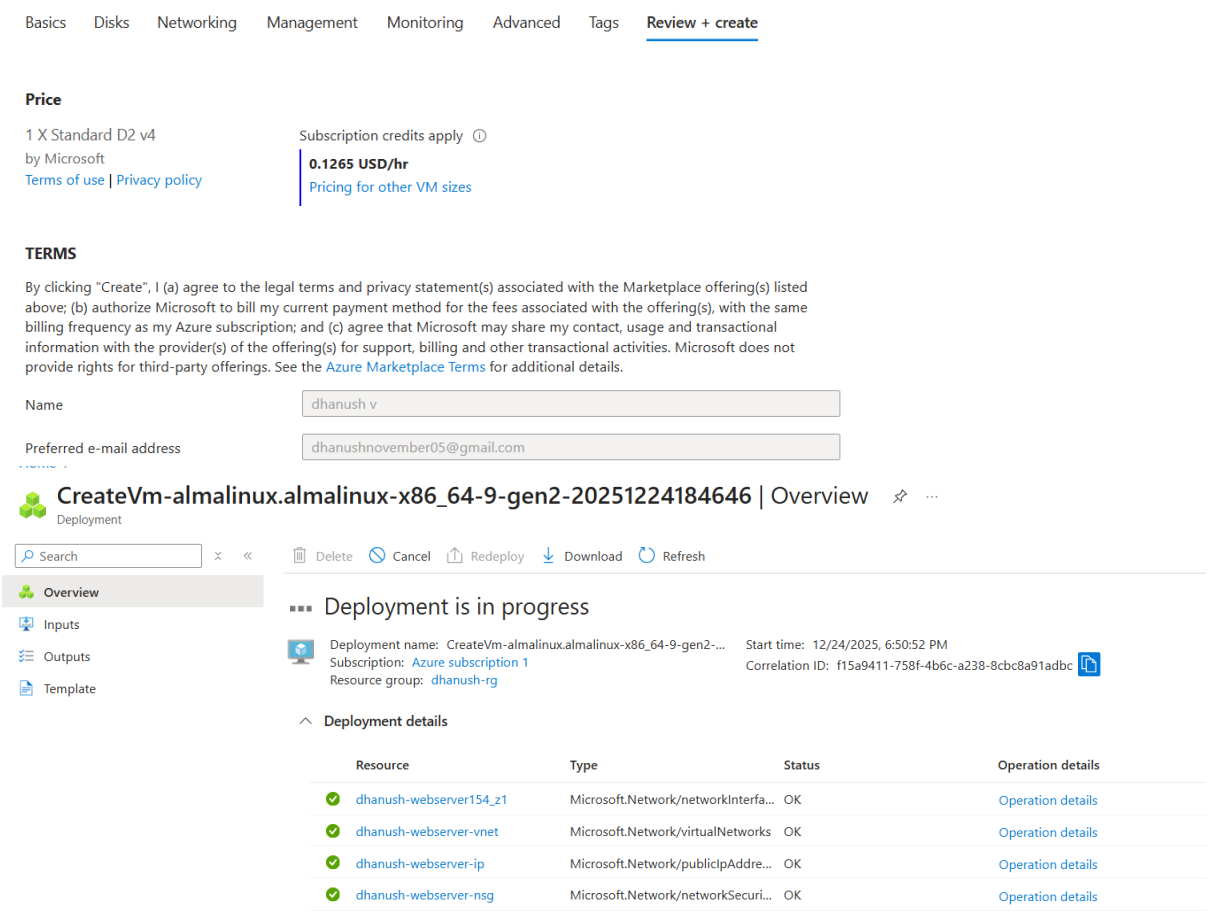
Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ:

- ( ) None
- (●) Allow selected ports

Select inbound ports *: HTTP (80), SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.



STEP 3 :Connect to VM

After deployment, note the Public IP.

Connect using SSH:

ssh -i <studentname>-key.pem azureuser@<public-ip>

```
C:\Users\DHANUSH>ssh -i dhanush-rg-key.pem azureuser@51.103.208.8
```

STEP4 : Install and Configure Apache HTTPD

```
[azureuser@dhanush-webserver ~]$ sudo dnf update -y
AlmaLinux 9 - AppStream
AlmaLinux 9 - BaseOS
```

```
[azureuser@dhanush-webserver ~]$ sudo dnf update -y
AlmaLinux 9 - AppStream
AlmaLinux 9 - BaseOS
```

```
[azureuser@dhanush-webserver ~]$ sudo dnf install -y httpd
Last metadata expiration check: 0:02:07 ago on Wed 24 Dec 2025 01:26:32 PM UTC.
Dependencies resolved.
================================================================================
 Package                 Architecture        Version                 Repository
```

```
[azureuser@dhanush-webserver ~]$ sudo systemctl start httpd
[azureuser@dhanush-webserver ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[azureuser@dhanush-webserver ~]$ cd /var/www/html/
```

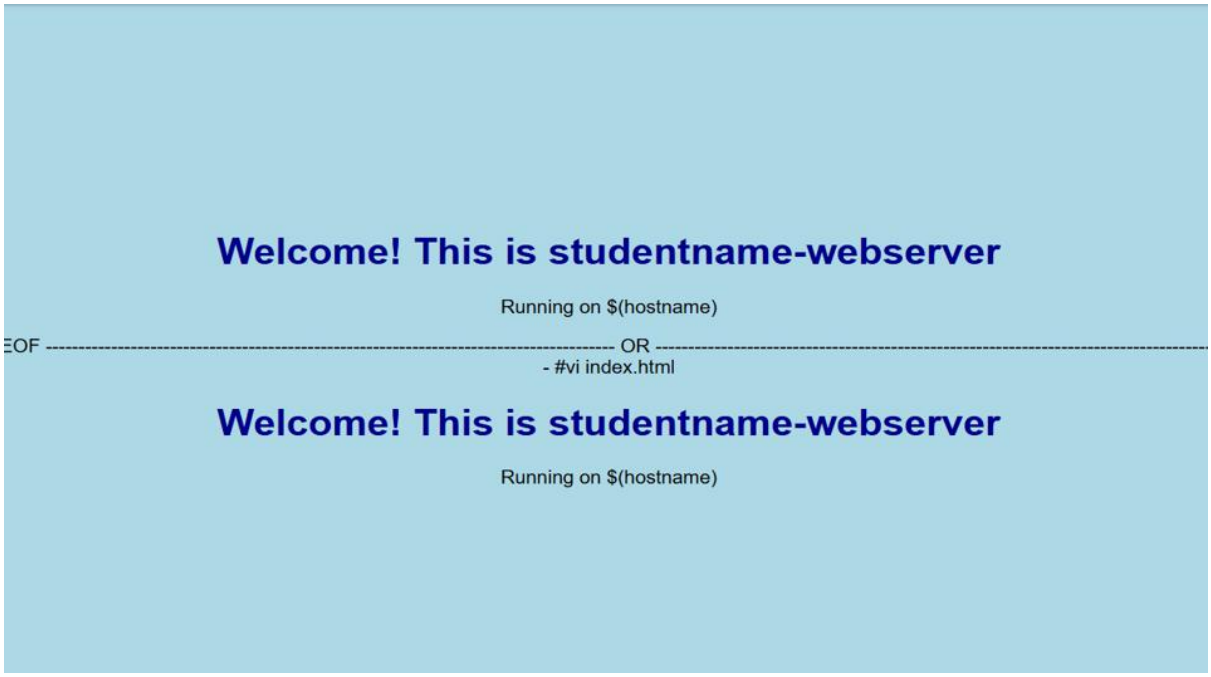
```
[azureuser@dhanush-webserver html]$ vi index.html
```

Enter the code in this vi index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>My Azure Webserver</title>
  <style>
    body {
      background-color: lightblue;
      font-family: Arial;
      text-align: center;
      margin-top: 20%;
    }
    h1 { color: darkblue; }
  </style>
</head>
<body>
  <h1>Welcome! This is studentname-webserver</h1>
  <p>Running on $(hostname)</p>
</body>
</html>
EOF
--------------------------------------------------------------------------------
OR
--------------------------------------------------------------------------------
#vi index.html
<!DOCTYPE html>
<html>
<head>
  <title>My Azure Webserver</title>
  <style>
    body {
      background-color: lightblue;
      font-family: Arial;
      text-align: center;
      margin-top: 20%;
    }
    h1 { color: darkblue; }
  </style>
</head>
<body>
  <h1>Welcome! This is studentname-webserver</h1>
```

Copy the public ip paste it in the browser

Output :



Update packages

sudo dnf update -y

Install Apache HTTP server

sudo dnf install -y httpd

Start and enable service

sudo systemctl start httpd

sudo systemctl enable httpd

Create a sample HTML page

#sudo tee /var/www/html/index.html > /dev/null <<EOF

```
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